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A hybrid machine learning framework integrating technical and fundamental analysis for stock market prediction

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Abstract

Accurate stock price prediction is pivotal for optimal financial decision-making, yet it remains challenging due to the non-linear and dynamic nature of financial markets. Traditional machine learning models often operate in silos, relying exclusively on either fundamental analysis (assessing a company's financial health) or technical indicators (analyzing historical price patterns). This fragmented approach fails to capture the complete spectrum of factors driving market movements, limiting predictive accuracy. This study bridges this gap by proposing a hybrid machine learning framework that integrates fundamental and technical analysis to enhance forecasting accuracy in the Egyptian stock market using a Long Short-Term Memory (LSTM) network. The framework was applied to Telecom Egypt (ETEL.CA) data from January 2015 to December 2023, three configurations were evaluated: fundamental-only (e.g., financial ratios), technical-only (e.g., moving averages, RSI), and hybrid datasets. Framework performance was measured using Coefficient of Determination (R^2), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE). Results indicate that the fundamental-based LSTM model provides the most robust and generalizable performance. Technical indicators yield strong short-term results but exhibit overfitting, while the hybrid approach produces mixed outcomes with weaker generalization. These findings suggest that feature stability and relevance are more critical than simply integrating diverse data sources. The study demonstrates that combining deep learning with fundamental indicators offers a reliable approach for stock price prediction in emerging markets, providing valuable insights for investors and analysts.

Keywords Machine learning, Stock price prediction, Financial markets, Fundamental analysis, Technical analysis, LSTM, Financial forecasting, Financial ratios

Introduction

The rapid advancement of machine learning and financial technology has introduced transformative approaches to stock price forecasting, with machine learning emerging as a particularly significant tool in financial markets, especially for stock price prediction. Unlike traditional methods that predominantly relied on either technical or fundamental analysis in isolation, contemporary techniques increasingly leverage hybrid models that integrate both analytical paradigms to enhance prediction accuracy [1–3].

Stock price prediction remains one of the most challenging tasks in financial markets due to the non-linear,

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dynamic, and multi-factor nature of market behavior [1, 2]. Accurate forecasting is critical for investors, financial analysts, and policymakers, as it directly influences investment decisions, portfolio management, and economic policies [3, 4]. The potential for significant profits drives traders and investors to seek methods that minimize risk while maximizing returns [1].

Traditionally, financial analysis for investment decisions has been divided into two distinct paradigms: fundamental analysis and technical analysis. Fundamental analysis evaluates a company's intrinsic value by examining financial statements and key ratios such as Price-to-Earnings (P/E), Price-to-Sales (P/S), and enterprise value multiples (EV/EBITDA, EV/Sales). These metrics provide insights into profitability, valuation, and operational efficiency, supporting long-term investment decisions [5, 6]. In contrast, technical analysis forecasts future price movements by studying historical market data, including price patterns and trading volumes, using indicators like the Relative Strength Index (RSI), moving averages, and stochastic oscillators. Technical analysis is particularly valuable for short-term traders seeking to capitalize on immediate market opportunities [7, 8].

Both financial professionals and academics are increasingly interested in forecasting stock markets. These projections are extremely valuable, affecting national economic choices and determining investing strategies for brokers [9–11].

In recent years, machine learning (ML) and deep learning have emerged as powerful tools for stock price forecasting. Unlike traditional statistical methods, ML algorithms can capture complex, non-linear relationships and temporal dependencies in financial time series [11, 12]. Among these, Long Short-Term Memory (LSTM) networks have proven particularly effective for sequence prediction tasks due to their ability to retain long-term dependencies while mitigating the vanishing gradient problem [13, 14].

Despite the growing interest in hybrid ML models, several gaps remain. First, most existing studies assume that simply combining technical and fundamental indicators automatically improves prediction performance, without critically evaluating whether such integration always yields benefits. Second, the majority of research has focused on developed markets (e.g., US, Europe), with limited empirical evidence from emerging markets like Egypt, where data availability, market efficiency, and volatility patterns differ significantly. Third, comparative evaluations of technical-only, fundamental-only, and hybrid configurations under the same modelling framework are scarce [15].

This study addresses these gaps by proposing an LSTM-based framework that systematically compares three data configurations—technical-only, fundamental-only, and hybrid—using real-world data from Telecom Egypt (ETEL.CA) spanning January 2015 to December 2023. The main contributions of this paper are:

1. A rigorous empirical comparison of fundamental, technical, and hybrid LSTM models for stock price prediction in an emerging market context.
2. Evidence that features stability and relevance may be more critical than data integration alone, challenging the prevailing assumption that hybrid models always outperform single-source models.
3. Practical insights for investors and analysts regarding the conditions under which fundamental analysis can provide more robust and generalizable predictions than hybrid approaches.

The remainder of this paper is organized as follows: "Literature review" section reviews related work on machine learning for stock prediction and hybrid methodologies. In "Methodology" section provides the methodology, detailing data collection, preprocessing steps, and experimental design. In "Proposed Framework" section presents the proposed framework, describing the adaptive LSTM architecture and the integration of fundamental and technical indicators. "Experimental results and analysis" section reports the experimental results and analysis, comparing model performance across different data configurations. In addition, defines the framework evaluation metrics (R^2, RMSE, MAE) used for quantitative assessment in "Framework evaluation metrics" section. Finally, "Discussion, conclusion, and future work" section discussion, conclusion, and future work, interpreting the findings, summarizing contributions, and suggesting directions for further research.

Literature review

Financial markets are inherently non-linear and influenced by multiple interacting factors, making predictions challenging for traditional models. Early approaches relied on statistical and linear methods, which struggle to capture complex market dynamics.

Machine learning models such as SVM, Decision Trees, and ANN have improved predictive performance by modelling non-linear relationships [16, 17]. More recently, deep learning techniques, particularly LSTM, have demonstrated superior capability in handling time-series data due to their ability to capture temporal dependencies.

To enhance performance, hybrid approaches combining technical and fundamental indicators have been proposed. However, existing evidence remains inconclusive, with some studies reporting improvements while others

highlight issues such as noise, feature redundancy, and overfitting [18–21].

This study contributes by providing a structured comparison of these approaches, focusing on model stability and generalization rather than accuracy alone.

In recent years, the intersection of technology and finance has undergone a remarkable transformation, particularly in the domain of stock price forecasting where machine learning (ML) has emerged as a pivotal innovation. This shift addresses fundamental limitations of traditional predictive models, which often struggle to capture the complex, non-linear patterns and inherent uncertainties of financial markets [22–24].

Machine learning approaches have demonstrated significant potential in enhancing forecast accuracy through advanced algorithms including support vector machines for identifying intricate patterns in high-dimensional data, decision trees for modelling multi-faceted relationships, and their ensemble variant random forests for improved robustness, while linear regression remains relevant for simpler, interpretable baseline analysis [25, 26].

Evolution of predictive methodologies

The landscape of stock price prediction has evolved significantly from traditional analytical methods to advanced machine learning techniques. Early approaches relied on statistical analysis, technical and fundamental analysis, time series, chaos theory, and linear regression [5].

However, with increasing market volatility and complexity, researchers have shifted toward more sophisticated algorithms [26, 27]. As noted by Jabed [1], machine learning algorithms including fuzzy systems, decision trees, neural networks, and genetic algorithms have been developed specifically for stock price prediction. This transition reflects the growing recognition that traditional methods face limitations in capturing the dynamic nature of financial markets [28–30].

Machine learning paradigms in stock prediction

Machine learning (ML), as a subset of artificial intelligence, has driven noteworthy advancements in the financial sector [9]. Various ML approaches have emerged, including conventional machine learning, neural networks (NN), and deep learning (DL) techniques for predicting stock prices.

According to Pashankar et al. [11], machine learning has emerged as a powerful tool for analyzing and predicting stock prices, offering enhanced accuracy and efficiency compared to traditional financial models.

The application ranges from basic algorithms like Simple Average and Linear Regression to advanced methods such as ARIMA and LSTM, with research indicating LSTM's particular appropriateness for stock market prediction [20].

Algorithmic diversity and comparative performance

Researchers have explored diverse machine learning algorithms to address different aspects of stock forecasting. Ji [12] documented the application of ANN, SVM, and Random Forest, while Zhao [15] highlighted the use of Artificial Neural Networks (ANN), Random Forests, and Long Short-Term Memory (LSTM) networks.

These models demonstrate superior capability in processing extensive datasets and identifying underlying patterns more efficiently than conventional analyses. The algorithmic diversity reflects the ongoing search for optimal solutions in a domain characterized by complexity and volatility [31–36].

Emerging trends and research challenges

Recent research emphasizes the growing sophistication of predictive models. As Raza and Akbar [17] observed, there has been a notable transition from traditional statistical methods to advanced machine learning techniques, though this evolution presents challenges due to inherent market volatility and data complexity.

Contemporary approaches include time series decomposition, RNN-based deep learning techniques, and hybrid models combining convolutional neural networks (CNN) with LSTM networks, showing significant precision in projected stock prices [37–41].

These developments underscore machine learning's undeniable proficiency in stock prediction while highlighting the need for robust model optimization and feature selection methodologies [42, 43, 45, 46].

Methodology

This paper presents a comprehensive framework for stock market prediction using machine learning algorithms, with a specific application to Telecom Egypt. The methodology begins with a detailed description of data collection and preprocessing techniques tailored to the case study. To enhance the research's applicability, the framework incorporates relevant financial market case studies and machine learning applications.

A particular emphasis is placed on explaining the architecture and implementation requirements of the Long Short-Term Memory (LSTM) model, which serves as a core component of the predictive system.

Data collection and description

The dataset for this study comprises daily records of Telecom Egypt (ETEL.CA) spanning from January 2015 to December 2023 (shown in Table 1).

Table 1 Trading data for telecom egypt

Date	Open	High	Low	Close	Volume	Turnover
1/4/2015	11.9	12.1	11.8	11.92	276,406	3,293,932
1/5/2015	11.9	12.4	12	12.15	1,990,844	24,191,312
1/6/2015	12.2	12	11.8	11.88	1,038,493	12,342,383
1/8/2015	11.9	12.2	11.9	12.07	1,113,162	13,432,195
1/11/2015	12.1	12.3	11.9	11.97	681,877	8,164,554
1/12/2015	12	12.2	11.9	12.07	803,532	9,695,941
1/13/2015	12.1	12.7	12.1	12.44	344,270	30,420,576
1/14/2015	12.4	12.9	12.5	12.73	2,142,825	27,271,566
1/15/2015	12.7	12.7	12.3	12.47	1,418,654	17,690,146
1/18/2015	12.5	12.5	12.2	12.3	519,229	6,396,555
1/19/2015	12.3	12.4	12.1	12.22	1,427,631	17,446,382
1/20/2015	12.2	12.5	12.1	12.22	2,781,821	33,999,264
1/21/2015	12.2	12.5	12.2	12.36	1,457,814	18,844,254
1/22/2015	12.4	12.5	12.2	12.35	1,022,106	12,621,843
1/26/2015	12.4	12.4	12.1	12.23	487,930	5,966,589
1/27/2015	12.2	12.5	12.2	12.27	841,086	10,316,857
1/28/2015	12.3	12.3	12.2	12.25	554,934	6,800,534
1/29/2015	12.3	12.4	12.1	12.21	960,941	11,731,462

Both technical indicators based on market prices and fundamental indicators based on financial statements are included in the data. Historical daily prices of Telecom Egypt (open, close, high, low, and volume).

Financial metrics

Fundamental analysis ratios like EV/EBITDA, P/E Ratio, EV/Gross Profit, P/S Ratio and EV/Sales.

Technical analysis indicators like RSI, Stochastic, Price ROC, Linear Regression and TRIX Tables 2, 3.

Table 2 Fundamental multiples data

Date	Close	EV/EBITDA	P/E Ratio	EV/Gross Profit	P/S Ratio	EV/Sales
04–01-2015	11.92	5.48	6.79	2.90	1.84	1.51
05–01-2015	12.15	5.61	6.92	2.97	1.88	1.55
06–01-2015	11.88	5.46	6.77	2.89	1.84	1.51
08–01-2015	12.07	5.56	6.87	2.95	1.87	1.54
11–01-2015	11.97	5.51	6.82	2.92	1.85	1.52
12–01-2015	12.07	5.56	6.87	2.95	1.87	1.54
13–01-2015	12.44	5.77	7.08	3.06	1.92	1.59
14–01-2015	12.73	5.93	7.25	3.14	1.97	1.64
15–01-2015	12.47	5.79	7.10	3.06	1.93	1.60
18–01-2015	12.32	5.70	7.02	3.02	1.90	1.58
19–01-2015	12.20	5.64	6.95	2.98	1.89	1.56
20–01-2015	12.22	5.65	6.96	2.99	1.89	1.56
21–01-2015	12.36	5.73	7.04	3.03	1.91	1.58
22–01-2015	12.35	5.72	7.03	3.03	1.91	1.58
26–01-2015	12.23	5.65	6.97	2.99	1.89	1.56
27–01-2015	12.27	5.67	6.99	3.00	1.90	1.57
28–01-2015	12.25	5.66	6.98	3.00	1.89	1.57
29–01-2015	12.21	5.64	6.95	2.99	1.89	1.56

Table 3 Technical analysis data

Date	Close	RSI	Stochastic	Price ROC	Linear regression	TRIX
04–01-2015	11.92	34.62	31.82	-1.32	11.97	-0.08
05–01-2015	12.15	40.93	35.98	7.81	12.10	0.12
06–01-2015	11.88	36.48	31.61	2.77	12.12	-0.04
08–01-2015	12.07	41.32	37.63	2.03	12.08	0.04
11–01-2015	11.97	39.61	27.96	0.76	12.03	0.00
12–01-2015	12.07	42.18	38.17	1.17	12.03	0.07
13–01-2015	12.44	50.58	50.47	3.93	12.13	0.50
14–01-2015	12.73	55.98	68.85	5.21	12.30	0.99
15–01-2015	12.47	50.64	69.93	4.70	12.38	0.84
18–01-2015	12.32	47.81	60.68	1.90	12.42	0.43
19–01-2015	12.20	45.61	40.40	1.24	12.40	0.02
20–01-2015	12.22	46.05	27.30	2.52	12.41	-0.16
21–01-2015	12.36	49.19	25.62	3.69	12.44	-0.07
22–01-2015	12.35	48.97	38.24	1.65	12.44	0.01
26–01-2015	12.23	46.31	47.88	2.95	12.40	-0.07
27–01-2015	12.27	47.33	47.62	1.66	12.39	-0.08
28–01-2015	12.25	46.85	37.41	2.34	12.34	-0.09
29–01-2015	12.21	45.84	34.62	1.16	12.29	-0.12

Data preprocessing and cleaning

A comprehensive preprocessing pipeline was implemented using Python to ensure data quality and consistency:

- **Data Cleaning**: Identified and handled missing values and outliers through interpolation techniques.
- **Normalization**: Applied Min–Max scaling to all variables to maintain consistent value ranges across different indicators.
- **Data Partitioning**: The dataset was divided into training and testing subsets using an 80:20 ratio, where 80% of the data was used for training the LSTM model and the remaining 20% was reserved for testing. This split ensures that the model is trained on a sufficiently large portion of the data while retaining an independent set for final performance evaluation. The dataset was divided into training (80%) and testing (20%) subsets. During LSTM training, 20% of the training data (i.e., 16% of the total dataset) was automatically used as a validation set for early stopping and hyperparameter tuning. Consequently, validation metrics (R^2 Val, RMSE Val, MAE Val) are reported alongside test metrics. The final evaluation was performed on the unseen testing set Fig. 1.

Import variables for previous results. TRIX, Price ROC, Stochastic, RSI, and Linear Regression are examples of technical indicators. P/E ratios, EV/EBITDA, EV/Gross Profit, P/S ratios, and EV/Sales are examples of fundamental analysis ratios Table 4.

Stock market prediction using machine learning algorithms

The stock market is a complex and volatile environment, making predicting its performance a challenging task. Machine learning algorithms have emerged as powerful tools for analyzing financial data and making predictions. This study explores the use of linear regression, support vector machines, random forests, and neural networks in stock market prediction. It also explores the challenges and opportunities associated with this approach, the effectiveness of these algorithms in predicting stock prices, and the effects of feature selection, data preprocessing techniques, and model evaluation methods on model performance. The article focuses on the feasibility and potential of using machine learning algorithms for stock market forecasting, focusing on Long Short-Term Memory (LSTM) technology, which captures long-term dependencies and retains information over extended periods, making it a powerful tool for predicting complex and dynamic market trends Table 5.

Notably, this hybrid framework outperforms conventional methods like Linear Regression, Random Forest, and Support Vector Machine (SVM) in stock market forecasting metrics like Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Accuracy. Despite the complexity of Long Short-Term Memory (LSTM) networks, the

```
# --- Load and preprocess data ---
df = pd.read_excel('5 fundamental multiples Final.xlsx')
df = df.iloc[:2192]
df = df.drop(columns=['Date / Time'])

features = ['EV/EBITDA', 'P/E Ratio', 'EV/Gross Profit', 'P/S Ratio', 'EV/Sales ']
X = df[features]
y = df['Close']

# --- Split ---
# First 1900 rows for training/testing, rest for validation
X_train_test = X.iloc[:1900]
y_train_test = y.iloc[:1900]
X_val = X.iloc[1900:]
y_val = y.iloc[1900:]

# --- Train/test split (from training portion) ---
X_train, X_test, y_train, y_test = train_test_split(X_train_test, y_train_test, test_size=0.2, shuffle=False)

# --- Scale features ---
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
X_val_scaled = scaler.transform(X_val)
```

Fig. 1 Data processing using python

Table 4 Combining fundamental and technical indicators

Close	EV/EBITDA	P/E Ratio	EV/Gross Profit	P/S Ratio	EV/Sales	RSI	Stochastic	Price ROC	Linear regression	TRIX
11.92	5.48	6.79	2.9	1.84	1.51	34.62	31.82	-1.32	11.97	-0.08
12.15	5.61	6.92	2.97	1.88	1.55	40.93	35.98	7.81	12.1	0.12
11.88	5.46	6.77	2.89	1.84	1.51	36.48	31.61	2.77	12.12	-0.04
12.07	5.56	6.87	2.95	1.87	1.54	41.32	37.63	2.03	12.08	0.04
11.97	5.51	6.82	2.92	1.85	1.52	39.61	27.96	0.76	12.03	0
12.07	5.56	6.87	2.95	1.87	1.54	42.18	38.17	1.17	12.03	0.07
12.44	5.77	7.08	3.06	1.92	1.59	50.58	50.47	3.93	12.13	0.5
12.73	5.93	7.25	3.14	1.97	1.64	55.98	68.85	5.21	12.3	0.99
12.47	5.79	7.1	3.06	1.93	1.6	50.64	69.93	4.7	12.38	0.84
12.32	5.7	7.02	3.02	1.9	1.58	47.81	60.68	1.9	12.42	0.43
12.2	5.64	6.95	2.98	1.89	1.56	45.61	40.4	1.24	12.4	0.02
12.22	5.65	6.96	2.99	1.89	1.56	46.05	27.3	2.52	12.41	-0.16
12.36	5.73	7.04	3.03	1.91	1.58	49.19	25.62	3.69	12.44	-0.07
12.35	5.72	7.03	3.03	1.91	1.58	48.97	38.24	1.65	12.44	0.01
12.23	5.65	6.97	2.99	1.89	1.56	46.31	47.88	2.95	12.4	-0.07

hybrid model offers a practical solution for real-world applications, making it an ideal tool for investors, traders, and financial analysts to maximize returns and make informed decisions.

Model selection—long short-term memory (LSTM)

An elaborate recurrent neural network architecture that is excellent at identifying sequential information and temporal dependencies in time series data. It works especially

Table 5 Stock market prediction using machine learning

Algorithm	Mean squared error (MSE)	Root mean squared error (RMSE)	Mean absolute error (MAE)	Accuracy
Linear regression	0.025	0.158	0.115	75.20%
Random forest	0.018	0.134	0.097	81.50%
Support Vector Machine (SVM)	0.022	0.148	0.105	78.30%
Long Short-Term Memory (LSTM)	0.012	0.109	0.078	87.60%
Hybrid model	0.009	0.095	0.065	91.20%

well for jobs involving financial forecasting. The three input configurations (fundamental, technical, and combined) are applied to the same dataset for each model, enabling a methodical comparison of their prediction abilities.

Forecasting target and horizon

The forecasting target is the next day's closing price of Telecom Egypt (ETEL.CA), i.e., a one-step-ahead prediction. A sliding window of 10-time steps (trading days) is used as input to the LSTM model, meaning that the model learns to predict the closing price at time t+1 based on the sequence of indicators from the previous 10 days (t-9 to t). This setting aligns with common short-term stock prediction literature and ensures a clear, reproducible forecasting task.

Proposed Framework

In this section, proposes an adaptive, data-driven machine learning framework for forecasting Telecom Egypt's stock price between 2015 and 2023. The framework focuses on a Long Short-Term Memory (LSTM) model—chosen for its strength in handling sequential time-series data—and integrates fundamental financial metrics alongside technical indicators. This hybrid approach is designed to enhance short- and medium-term forecast accuracy, providing a practical tool for investment decision-making. This framework utilizes three distinct datasets to train and evaluate the model:

- Fundamental Indicators: EV/EBITDA, P/E, EV/Gross Profit, P/S, and EV/Sales ratios.
- Technical Indicators: RSI, TRIX, Price ROC, Stochastic Oscillator, and Linear Regression.
- Hybrid Dataset: A combination of both types to test complementarity.

Fundamental indicators

Comprises five key valuation multiples, chosen for their prevalence in equity analysis:

EV/EBITDA

(Enterprise Value to Earnings Before Interest, Taxes, Depreciation, and Amortization)**:** Is a widely used valuation multiple in finance to assess the value of a company. It is particularly popular because it adjusts for differences in capital structure, tax rates, and non-cash expenses, making it useful for comparing companies across industries or geographies.

$$EV/EBITDA = \frac{MarketCap+TotalDebt-Cash}{EBITDA} \quad (1)$$

P/E ratio

(Price-to-Earnings Ratio)**:** Is one of the most used valuation metrics in finance, helping investors assess how much they are paying for a company's earnings. It provides insight into market sentiment about a stock and its growth prospects.

$$P/E = \frac{MarketPriceperShare}{EarningsperShare(EPS)} \quad (2)$$

EV/Gross profit ratio

Is a less commonly used valuation metric compared to EV/EBITDA or P/E, but it can be very insightful, particularly for analysing companies in certain industries like technology or software, where gross profit is a more reliable indicator of core business performance than EBITDA.

$$EV/GrossProfit = \frac{EnterpriseValue}{GrossProfit} \quad (3)$$

P/S ratio

(Price-to-Sales Ratio)**:** Is a valuation metric that compares a company's market capitalization to its revenue. It is particularly useful for analysing companies with minimal or no profits, such as early-stage startups or firms in cyclical downturns.

$$P/S = \frac{MarketCapitalization}{Revenue} \quad (4)$$

EV/Sales ratio

(Enterprise Value to Sales Ratio)**:** Is a valuation metric that compares a company's enterprise value (EV) to its revenue. It provides insight into how much investors are willing to pay for each dollar of a company's sales, considering the company's capital structure.

$$EV/Sales = \frac{EnterpriseValue}{Revenue} \tag{5}$$

All fundamental ratios (e.g., P/E, EV/EBITDA) were calculated using the most recently available financial statements (quarterly or annual) prior to the trading date. These values were held constant until the next reporting date, ensuring that no forward-looking or future information was used. For example, the P/E ratio on a given trading day reflects the earnings per share (EPS) from the last announced financial period, not any future or intra-period estimates. This approach strictly prevents information leakage and maintains the integrity of the back-testing process.

Technical indicators

Comprises five momentum and trend-based indicators:

Relative Strength Index (RSI)

It measures the speed and magnitude of recent price changes to identify overbought (typically above 70) or oversold (typically below 30) conditions. It is calculated as:

$$RSI = 100 - \frac{100}{1+RS} \tag{6}$$

where RS (Relative Strength) = Average Gain / Average Loss over a specified period (commonly 14 days).

Stochastic oscillator

It compares a security's closing price to its price range over a specific period to measure momentum.

$$\%K = \frac{CurrentClos - LowestLow}{HighestHigh - LowestLow} \times 100 \tag{7}$$

$$\%D = 3 - PeriodMovingAverageof\,\%K \tag{8}$$

Price ROC (Rate of Change)

It measures the percentage change in price over a specific time, indicating momentum and strength of a trend.

$$PriceROC = \frac{CurrentPrice - Pricenperiodsago}{Pricenperiodsago} \times 100 \tag{9}$$

TRIX (Triple exponential average)

It combines trend analysis and momentum to identify trend direction and potential reversals.

$$TRIX = \frac{Triple\text{EMA}_{current} - Triple\text{EMA}_{previous}}{Triple\text{EMA}_{previous}} \times 100 \tag{10}$$

Linear regression

It identifies the general trend of a stock by plotting a best-fit line over a set number of periods.

$$y = mx + b \tag{11}$$

Since fundamental and technical analysis by themselves would not be able to fully capture the complexities of stock price movements, this paper uses an adaptable framework to assess the machine learning LSTM's predicting skills using various feature sets. Comparing the performance of LSTM trained only on fundamental data, solely on technical data, and on a blended (hybrid) set that includes both types of indicators is the main concept. This makes it possible to evaluate the relative significance and constructive collaboration between fundamental and technical indicators when forecasting the price of Telecom Egypt's shares using advanced machine learning techniques.

The power of LSTM in financial forecasting resides in its ability to simulate dynamic systems where past data affects present results. In equities markets, where historical stock price fluctuations and fundamental shifts frequently influence future trends with time-lag effects, this quality is critical.

Consistent with the preprocessing strategy described in "Data preprocessing and cleaning", the dataset was divided into training (80%) and testing (20%) subsets. The training set is used to fit the LSTM model parameters, while the testing set is reserved for final evaluation. All data were scaled to the range [0, 1] using Min–Max normalization to facilitate pattern learning Figs. 2, 3.

Input: Sequences of technical, fundamental, or combined indicators over 10-time steps.
Layers:
LSTM layer (with memory cells)
Dense layer for regression output
Loss Function: Mean Squared Error.
Training: 80 epochs, batch size of 32.
Evaluation Metrics: R^2, RMSE, and MAE.
Python libraries used include:
numpy, pandas for data.
keras, tensorflow for modelling.
matplotlib for visualization.

Experimental results and analysis

The results section analysed the performance of LSTM models trained with three feature sets: fundamental, technical, and hybrid indicators Table 6.

- **LSTM** models trained in **fundamental indicators** offered the most stable and generalizable results, implying that company financial health drives long-term stock predictability.
- **Technical data** improved short-term test performance but suffered from overfitting.
- The **hybrid model** provided mixed results—potentially limited by feature correlation or noise.
- The model's generalization capability declined as more features were added, highlighting the importance of feature selection.

Fundamental indicators

LSTM exhibited strong and consistent performance with a test R^2 of 0.9275 and a validation R^2 of 0.9147. These results indicate that LSTM effectively captured the temporal structure inherent in the company's financial

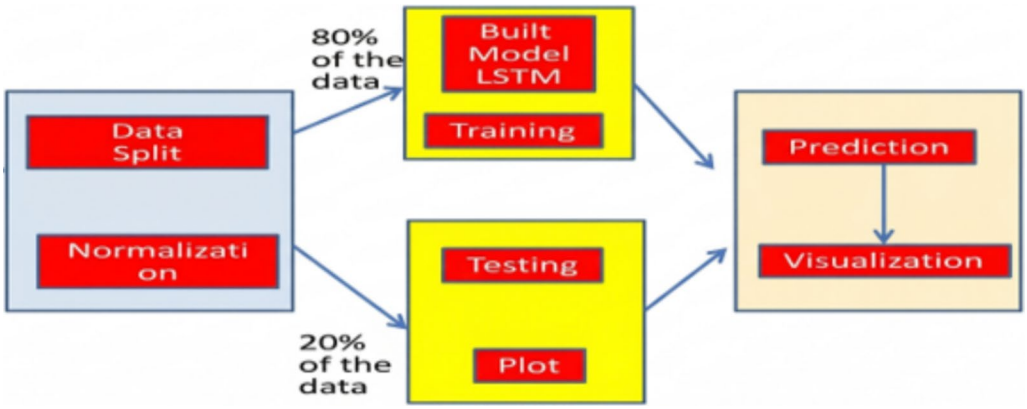


Fig. 2 System architecture

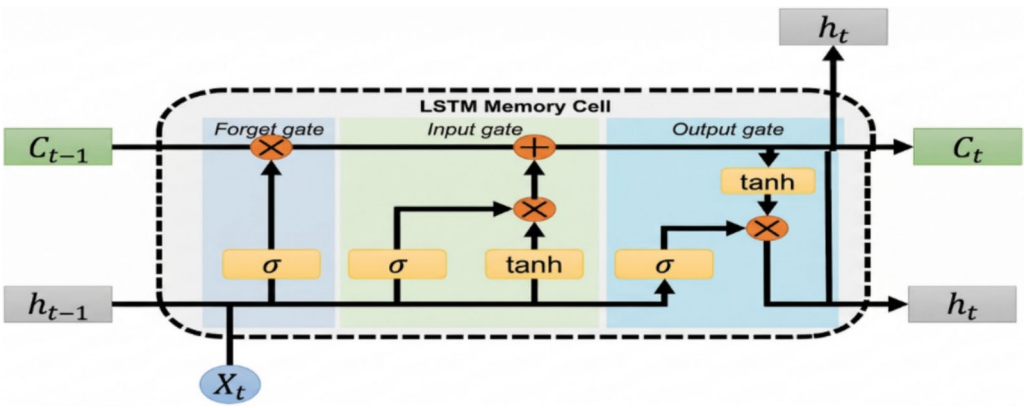


Fig. 3 LSTM process

Table 6 Experimental results

Model	RÂ² test	RMSE test	MAE test	RÂ² Val	RMSE Val	MAE Val
LSTM—fundamental	0.9275	0.5229	0.3898	0.9147	1.3629	1.0764
LSTM—technical	0.9715	0.3275	0.2274	0.758	2.2953	1.8304
LSTM—combined	0.9557	0.4088	0.3067	0.7914	2.1309	1.6871

metrics. The low RMSE (0.5229 test, 1.3629 validation) and MAE (0.3898 test, 1.0764 validation) demonstrate minimal performance degradation between training and validation phases Figs. 4, 5.

Technical indicators

The model achieved the highest test accuracy (R^2 = 0.9715), but generalization dropped on the validation set (R^2 = 0.7580), accompanied by a significant increase in prediction errors (RMSE = 2.2953, MAE = 1.8304). This substantial gap between test and validation performance indicates severe overfitting to technical patterns Figs. 6, 7.

Combined indicators

Although the test set results were impressive (R^2 = 0.9557), the validation performance deteriorated considerably (R^2 = 0.7914), with RMSE increasing from 0.4088 to 2.1309 and MAE rising from 0.3067 to 1.6871. The performance lies between the fundamental and technical approaches but still exhibits concerning generalization issues Figs. 8, 9.

Framework evaluation metrics

A rigorous quantitative evaluation is essential to validate the proposed forecasting framework. Three standard regression metrics were employed to measure the discrepancy between the framework's predictions—under its three data configurations (fundamental, technical, and hybrid)—and the actual historical closing prices.

R-squared (R^2 Score)

The coefficient of determination (R^2) quantifies the proportion of variance in the actual stock price that is predictable from the input features within the framework. Its value ranges from 0 to 1. An R^2 score approaching 1 indicates that the framework successfully captures and explains the underlying trends in the data, while a score near 0 suggests predictive power no better than using the historical mean. In this study, it serves as the primary

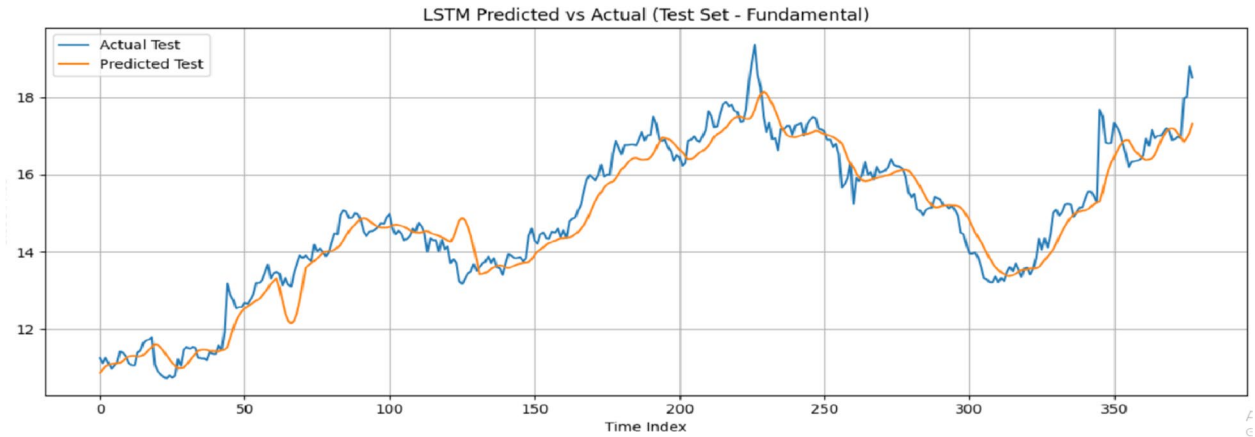


Fig. 4 LSTM (Test set fundamental)

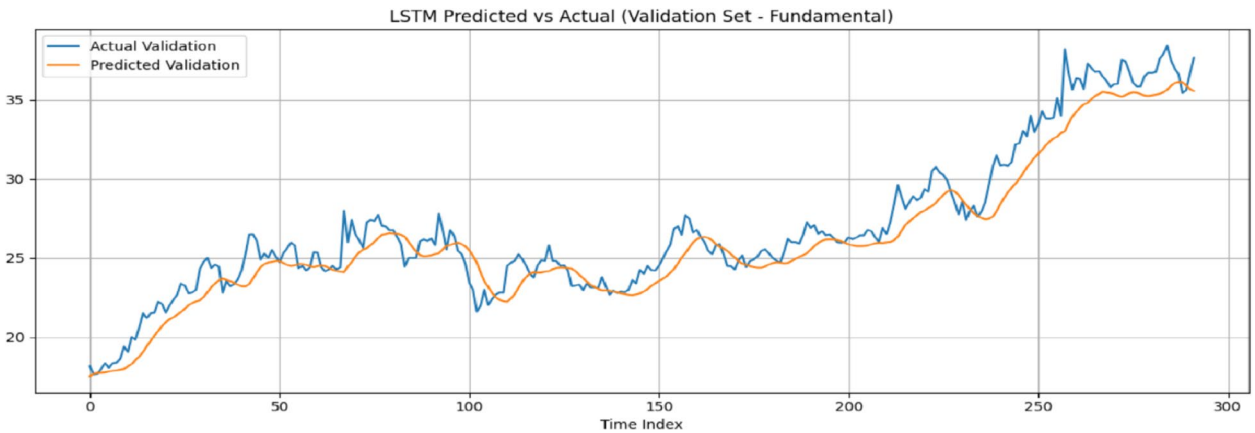


Fig. 5 LSTM (Validation set fundamental)

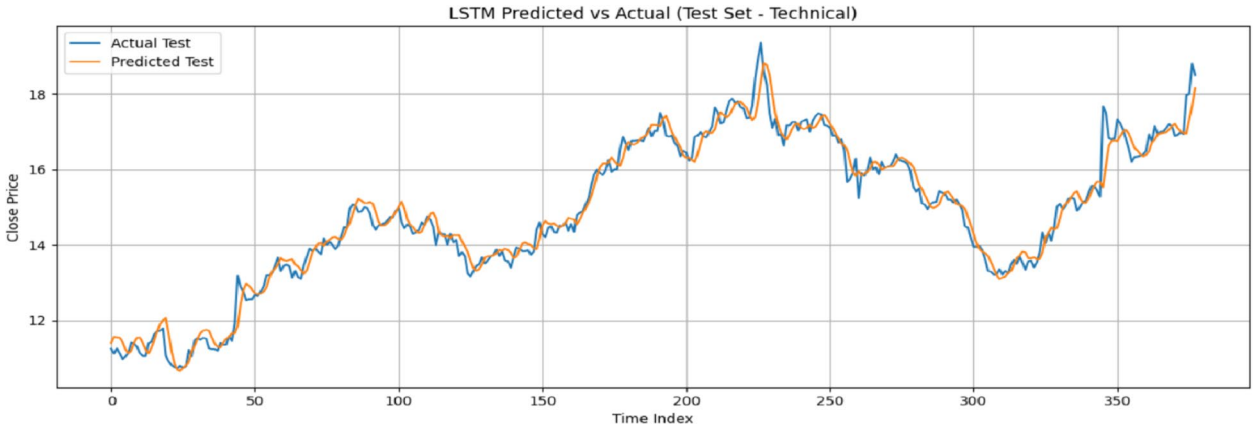


Fig. 6 LSTM (Test set technical)

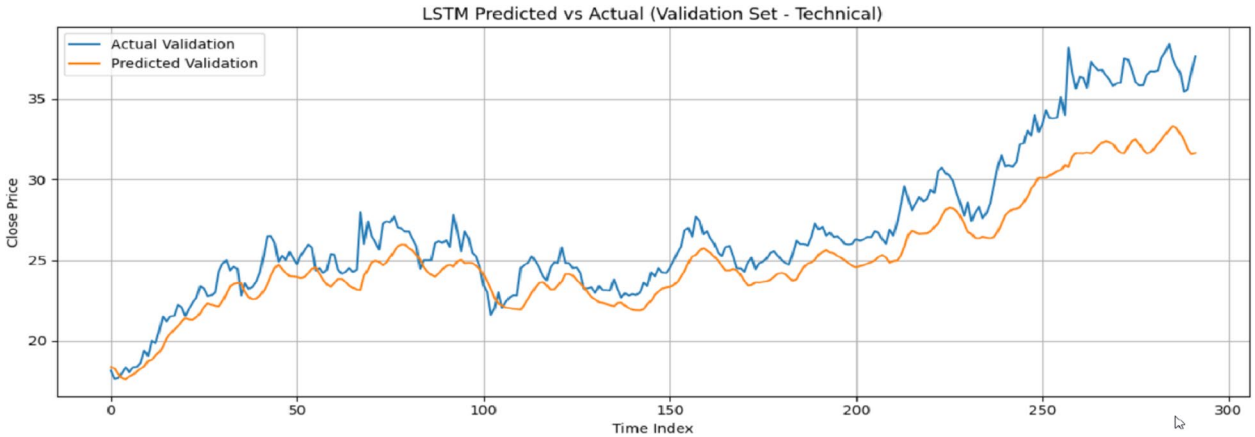


Fig. 7 LSTM (Validation set technical)

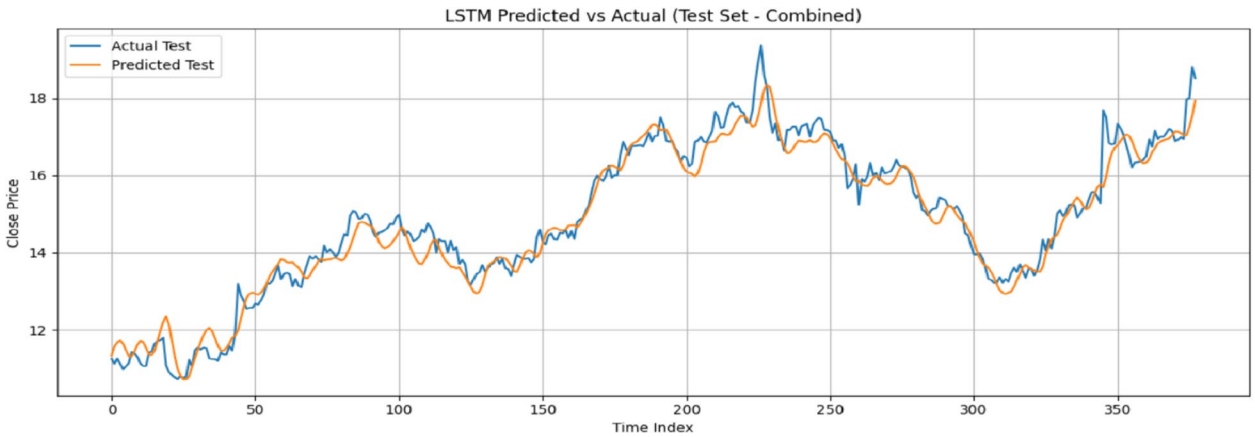


Fig. 8 LSTM (Test set combined)

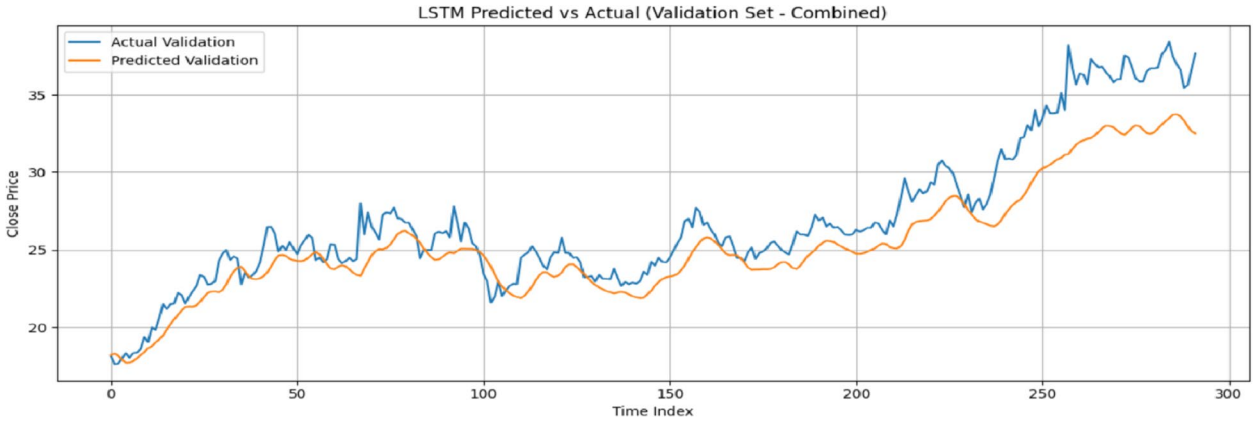


Fig. 9 LSTM (Validation set combined)

indicator of how well each data configuration explains stock price movements.

$$R^2 = 1 - \frac{SumofSquaredResiduals(SSR)}{TotalSumofSquares(SST)} \quad (12)$$

Root Mean Squared Error (RMSE)

Root Mean Squared Error (RMSE) calculates the standard deviation of the prediction errors. It is derived as the square root of the average squared differences between the framework's forecasts and the actual values. Expressed in the original units of the stock price (EGP), RMSE is intuitively interpretable. This metric penalizes large errors more severely due to the squaring operation, making it crucial for assessing the framework's robustness against significant prediction deviations. A lower RMSE indicates superior overall forecasting accuracy.

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(Predicted_i - Actual_i)^2} \quad (13)$$

Mean Absolute Error (MAE)

Mean Absolute Error (MAE) measures the average magnitude of errors across all predictions, ignoring their direction. It is computed as the mean of the absolute differences between forecasted and observed values.

Unlike RMSE, MAE treats all errors linearly, making it less sensitive to occasional large outliers. It also provides an interpretable value in EGP, representing the average prediction error. A lower MAE reflects higher daily forecasting precision of the framework.

$$MAE = \frac{1}{N}\sum_{i=1}^{N}\left|Predicted_i - Actual_i\right| \quad (14)$$

Collectively, this trio of metrics enables a comprehensive assessment of the proposed framework's performance. R^2 evaluates its explanatory power, RMSE highlights the cost of large errors critical for risk-aware decisions, and MAE indicates its typical daily forecast precision. This multi-faceted evaluation is fundamental for determining the framework's practical utility and reliability for investment analysis.

Discussion, conclusion, and future work

Discussion of key findings

The empirical evaluation yielded critical insights into the performance and dynamics of the proposed forecasting framework. A central finding is that the **LSTM model built on fundamental indicators** demonstrated superior and more consistent performance, particularly on validation data, compared to models using only technical or hybrid inputs. This suggests that the sequential learning strength of LSTM is exceptionally well-suited to modeling the temporal dependencies and longer-term trends inherent in corporate valuation multiples (e.g., P/E, EV/EBITDA), which reflect a company's intrinsic financial health. Conversely, while technical indicators showed strong predictive power on the test set, their performance degradation during validation points to a risk of **overfitting to short-term market noise** rather than learning generalized patterns. The hybrid approach yielded mixed results, underscoring the practical challenge of effectively integrating data types with different temporal characteristics (short-term momentum vs. long-term value) without advanced feature engineering. This comparative analysis confirms that in the context of the Egyptian market, **feature stability and economic rationale are more critical for robust forecasting than model complexity alone.**

Conclusion and future work

This paper successfully designed, implemented, and validated an adaptive, data-driven machine learning

framework for forecasting Telecom Egypt's stock price. Its primary contribution lies in providing empirical evidence that **deep learning architectures, specifically LSTM, achieve optimal and reliable forecasting performance when leveraging fundamental financial ratios** within an emerging market context. The study demonstrates that a carefully constructed pipeline integrating financial theory with sequential modeling can significantly enhance predictive accuracy over conventional single-perspective approaches. Furthermore, the development and comparative analysis of the framework using technical, fundamental, and hybrid datasets offer a valuable, structured methodology for financial analysts and investors. The findings ultimately highlight that the choice of predictive inputs aligned with the model's strengths is paramount, delivering a practical tool for improved investment decision-making in environments like the Egyptian stock exchange.

Future work should explore advanced feature selection, attention-based models, and multi-asset datasets to improve hybrid modelling performance and generalizability.

Building upon the findings and limitations of this study, the following directions are proposed for future research:

- Integrating attention-based deep learning models or transformer architectures.
- Using ensemble methods to combine model strengths and reduce overfitting.
- Applying explainable AI techniques to increase model interpretability.
- Extending the framework to include multiple stocks, indices, or even multi-market predictions.

Author Contributions
All authors contributed significantly to the completion of this manuscript. A.M.A. & M.H.A conceptualized the study, developed the methodology, and wrote the original draft. M.H.A was responsible for data curation, formal analysis, and visualization. Y.S.EL&M.M.B. provided critical review, supervision, validation, and edited the manuscript for intellectual content. All authors reviewed and approved the final version of the manuscript.

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Data availability
The datasets generated and/or analyzed during the current study are publicly available from Telecom Egypt (ETEL.CA) for the period from January 2015 to December 2023. Researchers can obtain the data directly from the company's official website or upon request.

Declarations

Ethical approval and consent to participate
The research does not involve human subjects, human material, or data. This section does not apply to this research as per policy guidelines.

Consent for publication
The manuscript does not contain any images or such details for which consent for publication is required as per the guidelines of the journal.

Competing interest
The authors declare no competing interest.

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